Early-Stage Diabetes Risk Prediction

Using Random Forest and XGBoost Classifiers

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Abstract

Diabetes is one of the most common health problems worldwide, and early detection is important to prevent serious complications. This project, Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers, aims to predict the risk of diabetes at an early stage using Machine Learning techniques. The system analyzes patient information such as age, gender, frequent urination, excessive thirst, sudden weight loss, weakness, family history, and other common symptoms to determine whether a person is likely to have diabetes. By providing early predictions, the system can help individuals seek timely medical advice and improve their overall health.

Abstract

The proposed system uses two popular Machine Learning algorithms, Random Forest and XGBoost, to classify patients as diabetic or non-diabetic. The dataset is first preprocessed by handling missing values, converting categorical data into numerical form, and splitting the data into training and testing sets. Both models are trained and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. The results are compared to identify the model that provides the best prediction performance. This project demonstrates how Machine Learning can support early disease detection and assist healthcare professionals in making faster and more accurate decisions.